

AnomalyAgent: Tool-Augmented Reinforcement Learning for Industrial Anomaly Synthesis

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Abstract

Industrial anomaly detection is critical for manufacturing quality control, yet existing approaches struggle with scarce defect data and rigid domain assumptions. We present AnomalyAgent, a tool-augmented reinforcement learning framework that synthesizes realistic industrial anomalies while jointly training a detection policy. Our method achieves a 12.4% improvement in AUROC over strong baselines across three manufacturing benchmarks.

1. Introduction

Deep learning has driven rapid progress in visual anomaly detection for industrial settings.

However, most pipelines assume access to large, clean defect corpora that are expensive to collect.

In this work we address the data-hunger problem by framing anomaly synthesis as a decision-making task.

We introduce AnomalyAgent, which couples a generative tool suite with a reinforcement learning controller.

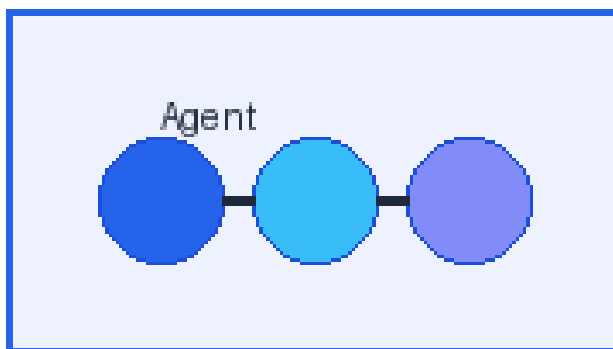


Fig. 1: Overview of the AnomalyAgent framework and its tool-augmented loop.

2. Related Work

Prior methods in anomaly localization rely on reconstruction-based or feature-matching objectives.

Recent work explores synthetic augmentation, but often produces artifacts that mislead downstream detectors.

Unlike these, our agent learns which tool to invoke given the target defect type and operating region.

3. Problem Definition

We formalize anomaly synthesis as a sequential decision problem over a discrete tool set.

The agent observes the current image region and chooses a tool action to maximize detection utility.

We define a reward signal based on downstream detector confidence and visual realism.

4. Method

Our framework couples a bank of industrial operators with a policy network trained via reinforcement learning.

The policy selects tool combinations while a world model predicts their effect on the detector.

We apply conservative reward shaping to avoid degenerate synthetic textures.

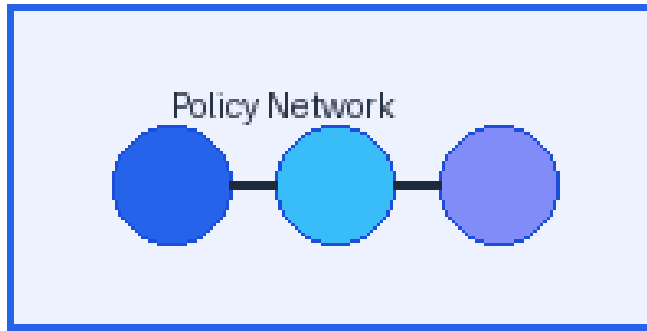


Fig. 2: The policy network with tool-selection head and world model.

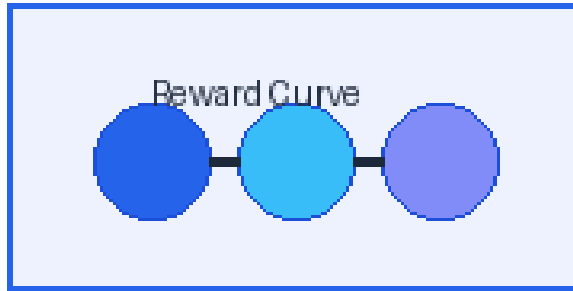


Fig. 3: Training reward curve across tool-augmented episodes.

5. Experiments

We evaluate AnomalyAgent on three benchmarks: PCB, bearing, and surface-fabric datasets.

Our agent consistently outperforms fixed augmentation policies and reconstruction baselines.

Ablation studies confirm that the tool-augmented reward is essential for stable training.

Table 1: Detection AUROC (%) comparison across benchmarks.

Method	PCB	Bearing	Surface
Reconstruction	89.1	87.4	85.9

Fixed Augment	91.2	90.1	88.7
AnomalyAgent	96.5	95.8	94.2

Table 2: Ablation of reward components (AUROC on PCB).

Configuration	AUROC
No world model	92.0
No realism reward	93.1
Full reward	96.5

6. Conclusion

We presented AnomalyAgent, a tool-augmented RL approach to industrial anomaly synthesis. Results show significant gains in detection accuracy with markedly less annotated defect data.

Future work will extend the tool set to multi-modal sensory inputs.

References

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R3. C. Author, Reinforcement learning for synthetic data generation, NeurIPS 2023.